HW #11

ALGEBRAIC NUMBER THEORY, GU4043; INSTRUCTOR: GYUJIN OH

Question 1. Let K be a number field. Let p be a rational prime that is unramified in K.

(1) Suppose that $K/\mathbb{Q}$ is Galois. Show that

$$\text{There exists } \alpha \in \mathcal{O}_K \text{ such that } p = |N_{K/\mathbb{Q}}(\alpha)| \Leftrightarrow p \text{ splits completely in } H_K.$$

(2) Suppose that $K/\mathbb{Q}$ is Galois. Suppose that K is **totally imaginary** (i.e., all archimedean primes of K are complex). Show that

$$\text{There exists } \alpha \in \mathcal{O}_K \text{ such that } p = N_{K/\mathbb{Q}}(\alpha) \Leftrightarrow p \text{ splits completely in } H_K.$$

(3) Show that, in general, only one direction of the equivalence in Question 1(1) is true.

(4) Find a counterexample to the other direction in Question 1(1) when $K/\mathbb{Q}$ is not Galois as follows.

(a) Let $K = \mathbb{Q}(\sqrt[3]{3})$. Show that $h_K = 1$.

Hint. By Example 2.15, you know $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{3}]$. Use Minkowski bound.

(b) Find p such that the prime ideal factorization of (p) in K is given by

$$(p) = \mathfrak{p}_1\mathfrak{p}_2, \quad f(\mathfrak{p}_1|p) = 1,\ f(\mathfrak{p}_2|p) = 2.$$

(c) Deduce that this gives a counterexample.

(5) Suppose that $K/\mathbb{Q}$ is Galois. Let H_K^+ be the **narrow Hilbert class field of** K; namely, $H_K^+ := K(\mathfrak{m})$ is the ray class field of modulus $\mathfrak{m} = \mathfrak{m}_f\mathfrak{m}_\infty$ where $\mathfrak{m}_f = (1)$ and $\mathfrak{m}_\infty$ is the product of all real embeddings of K. Show that (note there is only one direction!)

$$\text{There exists } \alpha \in \mathcal{O}_K \text{ such that } p = N_{K/\mathbb{Q}}(\alpha) \Leftarrow p \text{ splits completely in } H_K^+.$$

Proof. (1) We do this via a chain of equivalences.

There is $\alpha \in \mathcal{O}_K$ s.t. $p = |N_{K/\mathbb{Q}}(\alpha)| \Leftrightarrow$ There is a principal prime ideal $\mathfrak{p} \subset \mathcal{O}_K$ dividing p and $f(\mathfrak{p}|p) = 1$.

This is obvious as $N(\mathfrak{p}) = p^{f(\mathfrak{p}|p)}$ in general.

$$\text{There is a principal prime ideal } \mathfrak{p} \subset \mathcal{O}_K \text{ dividing } p \text{ and } f(\mathfrak{p}|p) = 1$$
$$\Leftrightarrow \text{There is a prime ideal } \mathfrak{p} \subset \mathcal{O}_K \text{ dividing } p \text{ s.t. } f(\mathfrak{p}|p) = 1 \text{ and } \mathfrak{p} \text{ splits completely in } H_K.$$

This is also obvious as $\mathfrak{p}$ is principal iff $\mathfrak{p}$ splits completely in H_K.

$$\text{There is a prime ideal } \mathfrak{p} \subset \mathcal{O}_K \text{ dividing } p \text{ s.t. } f(\mathfrak{p}|p) = 1 \text{ and } \mathfrak{p} \text{ splits completely in } H_K$$
$$\Leftrightarrow p \text{ splits completely in } H_K.$$

For $\Rightarrow$: as $K/\mathbb{Q}$ is Galois, p splits completely in H_K, and all other primes of K dividing p is a conjugate of $\mathfrak{p}$. Therefore, any prime of K dividing p, being a conjugate of $\mathfrak{p}$, splits completely in H_K. Therefore, p splits completely in H_K.

For $\Leftarrow$: obvoious.

(2) As K is totally imaginary, all archimedean primes of K are complex embeddings. In particular, the complex embeddings are paired in conjugates, and we have

$$N_{K/\mathbb{Q}}(\alpha) = \prod_{i=1}^{n} \sigma_i(\alpha)\overline{\sigma}_i(\alpha) = \prod_{i=1}^{n} |\sigma_i(\alpha)|^2 > 0,$$

where $[K : \mathbb{Q}] = 2n$ and $\{\sigma_1, \overline{\sigma}_1, \cdots, \sigma_n, \overline{\sigma}_n\}$ are the complex embeddings of K. Therefore, there is no need for the absolute value for $N_{K/\mathbb{Q}}(\alpha)$.

(3) In general, we only have

$$\text{There is a prime ideal } \mathfrak{p} \subset \mathcal{O}_K \text{ dividing } p \text{ s.t. } f(\mathfrak{p}|p) = 1 \text{ and } \mathfrak{p} \text{ splits completely in } H_K$$
$$\Leftarrow p \text{ splits completely in } H_K.$$

Therefore, only

$$\text{There exists } \alpha \in \mathcal{O}_K \text{ such that } p = |N_{K/\mathbb{Q}}(\alpha)| \Leftarrow p \text{ splits completely in } H_K$$

is true.

(4) Note that K has one real embedding and two complex embeddings (a pair of conjugate complex embeddings). Note also that $\mathrm{disc}(K) = D(1, \sqrt[3]{3}, \sqrt[3]{3}^2) = -27 \cdot 3^2 = -3^5$. So the Minkowski bound is

$$B_K = \frac{3!}{3^3}\frac{4}{\pi}\sqrt{3^5} = \frac{8\sqrt{3}}{\pi}.$$

Note that $8\sqrt{3} < 5\pi$, so $B_K \sim 4.\cdots$. Therefore, Cl_K is generated by ideal classes of prime ideals of norm $2, 3, 4$. As $\mathcal{O}_K = \mathbb{Z}[\sqrt[3]{3}]$, 2 and 3 factor as $X^3 - 3$ factor mod 2 and 3, so

$$(2) = \mathfrak{p}_2\mathfrak{p}_2', \quad \mathfrak{p}_2 = (2, \sqrt[3]{3} - 1),\ \mathfrak{p}_2' = (2, \sqrt[3]{3}^2 + \sqrt[3]{3} + 1),$$
$$(3) = \mathfrak{p}_3^3, \quad \mathfrak{p}_3 = (3, \sqrt[3]{3}).$$

This implies that Cl_K is generated by $[\mathfrak{p}_2], [\mathfrak{p}_2'], [\mathfrak{p}_3]$. On the other hand,

$$2 = (\sqrt[3]{3} - 1)(\sqrt[3]{3}^2 + \sqrt[3]{3} + 1) \rightsquigarrow \mathfrak{p}_2 = (\sqrt[3]{3} - 1), \mathfrak{p}_2' = (\sqrt[3]{3}^2 + \sqrt[3]{3} + 1),$$
$$\mathfrak{p}_3 = (\sqrt[3]{3}),$$

so that $[\mathfrak{p}_2], [\mathfrak{p}_2'], [\mathfrak{p}_3]$ are all trivial. Therefore, Cl_K is trivial.

Now $p = 2$ gives a counterexample to the $\Rightarrow$ direction of Question 1(1). Namely, $2 = |N_{K/\mathbb{Q}}(\alpha)|$ for some α; either because $(2) = \mathfrak{p}_2\mathfrak{p}_2'$ and $\mathfrak{p}_2, \mathfrak{p}_2'$ are principal with $f(\mathfrak{p}_2|2) = 1$, or more directly because

$$N_{K/\mathbb{Q}}(1 - \sqrt[3]{3}) = (1 - \sqrt[3]{3})(1 - \zeta_3\sqrt[3]{3})(1 - \zeta_3^2\sqrt[3]{3}) = 1^3 - 3 = -2.$$

On the other hand, as $f(\mathfrak{p}_2'|2) = 2$, 2 does not split completely in H_K.

(5) Suppose that p splits completely in H_K^+. This means that, firstly, p splits completely in K, and for each prime $\mathfrak{p}$ of K dividing p, $\mathfrak{p} = (\alpha)$ for $\alpha \in \mathcal{O}_K$ such that $\sigma(\alpha) > 0$ for every real embedding $\sigma : K \to \mathbb{R}$. Therefore,

$$N_{K/\mathbb{Q}}(\alpha) = \prod_{\sigma \text{ a real embedding of } K} \sigma(\alpha) \prod_{\{\sigma, \overline{\sigma}\} \text{ a conjugate pair of complex embeddings of } K} \sigma(\alpha)\overline{\sigma}(\alpha) > 0.$$

Therefore $p = N((\alpha)) = |N_{K/\mathbb{Q}}(\alpha)| = N_{K/\mathbb{Q}}(\alpha)$.

$\square$

Question 2. Let K be a number field. Suppose that a rational prime p is inert in H_K. Show that $h_K = 1$ (i.e., $H_K = K$).

Proof. Note that firstly p is inert in K as $K \subset H_K$. As $p\mathcal{O}_K$ is a principal prime ideal, $[p\mathcal{O}_K] = 1$ in Cl_K. Therefore, $p\mathcal{O}_K$ should split completely in H_K. As this stays inert, the only possibility is that $H_K = K$. $\square$

Question 3. We will study when a number is a fourth power (a **quartic residue**) modulo a rational prime, governed by what is called the **quartic reciprocity law**. Let p be a rational prime.

(1) Suppose that $p \equiv 3 \pmod 4$. Show that every quadratic residue mod p is a quartic residue mod p (i.e., if an integer a is a square mod p, it is a fourth power mod p).

Hint. For any integer b, either b or $-b$ is a quadratic residue mod p.

(2) Let $K = \mathbb{Q}(\sqrt{-1})$, which contains $\mu_4 = \{\pm 1, \pm\sqrt{-1}\}$. Note that $h_K = 1$, so that any ideal of $\mathcal{O}_K$ is principal. Let $\pi = 1 - \sqrt{-1}$, which generates the unique prime ideal $\mathfrak{p}_2 = (\pi)$ of K dividing 2.

Let $\mathfrak{p}$ be a prime ideal of $\mathcal{O}_K$ that divides a rational prime $p \equiv 1 \pmod 4$. Show that there exists a unique generator of $\mathfrak{p}$ congruent to $1 \pmod{\pi^3}$. We call such a generator a **primary number**.

Hint. Show that $\mathcal{O}_K^\times = \mu_4$.

(3) Let π_1, π_2 be primary numbers of K coprime to each other. If either π_1 or π_2 is further congruent to $1 \pmod 4$, show that

$$(\pi_1, \pi_2)_{\mathfrak{p}_2} = 1.$$

Hint. Adapt the proof of Exercise 5.6. Note that π is a uniformizer of $K_{\mathfrak{p}_2}$, and $4\mathcal{O}_K = \pi^4\mathcal{O}_K$.

(4) Consider the 4-th power residue symbol $\left(\frac{\cdot}{\cdot}\right) \in \mu_4$ (see Definition 5.46). Let π_1, π_2 be primary numbers of K coprime to each other. If either π_1 or π_2 is further congruent to $1 \pmod 4$, show that

$$\left(\frac{\pi_1}{\pi_2}\right) = \left(\frac{\pi_2}{\pi_1}\right).$$

(5) Let π_1, π_2 be primary numbers of K coprime to each other. If both π_1 nad π_2 are congruent to $3 + 2\sqrt{-1} \pmod 4$, show that

$$\left(\frac{\pi_1}{\pi_2}\right) = -\left(\frac{\pi_2}{\pi_1}\right).$$

Hint. First show that this follows from $(3 + 2\sqrt{-1}, 3 + 2\sqrt{-1})_{\mathfrak{p}_2} = -1$. Use Hilbert reciprocity law to show that $(3 + 2\sqrt{-1}, 3 + 2\sqrt{-1})_{\mathfrak{p}_2} = (3 + 2\sqrt{-1}, 3 + 2\sqrt{-1})_{\mathfrak{p}_{13}}$, where $\mathfrak{p}_{13} = (3 + 2\sqrt{-1})$. Compute $(3 + 2\sqrt{-1}, 3 + 2\sqrt{-1})_{\mathfrak{p}_{13}}$ using Theorem 5.44.

(6) (**Gauss's quartic reciprocity law**) Show that (4) and (5) can be neatly packaged into the following statement: if π_1, π_2 are primary numbers of K coprime to each other,

$$\left(\frac{\pi_1}{\pi_2}\right) = (-1)^{\frac{N_{K/\mathbb{Q}}(\pi_1)-1}{4} \cdot \frac{N_{K/\mathbb{Q}}(\pi_2)-1}{4}} \left(\frac{\pi_2}{\pi_1}\right).$$

Proof. (1) This is obvious because $(4, p-1) = 2$, so if $x = a^2 \bmod p$, then as $4k + (p-1)l = 2$ for some $k, l \in \mathbb{Z}$, $x \equiv a^2 \equiv a^{4k+(p-1)l} \equiv a^{4k} \pmod{p}$.

(2) Note that $\mathcal{O}_K = \mathbb{Z}[\sqrt{-1}]$, and for $x + y\sqrt{-1} \in \mathcal{O}_K$, $x + y\sqrt{-1} \in \mathcal{O}_K^\times$ if and only if $N_{K/\mathbb{Q}}(x + y\sqrt{-1}) = x^2 + y^2 = \pm 1$. As $x^2 + y^2 > 0$, this is only the case when $x^2 + y^2 = 1$, and this only happens if one of x, y is zero, and the other is ± 1. Therefore, $\mathcal{O}_K^\times = \mu_4$.

Note that $\pi^3 = (1 - \sqrt{-1})^2 = -2\sqrt{-1}(1 - \sqrt{-1}) = -2 - 2\sqrt{-1}$. For a rational prime $p \equiv 1 \pmod 4$, $p = x^2 + y^2$ for some $x, y \in \mathbb{Z}$, and there are four ways to take the generator of $(x + y\sqrt{-1})$, namely $\pm(x + y\sqrt{-1})$, $\pm(-y + x\sqrt{-1})$. Note that only one of x, y is even as p is odd, so without loss of generalities, let us assume that x is even and y is odd. Then $\pm(x + y\sqrt{-1})$ is not primary as y is odd. Also, $\pm(-y + x\sqrt{-1}) \equiv \pm(-x - y) \pmod{-2 - 2\sqrt{-1}}$ as x is even, and, as $x + y$ is odd, exactly one of $x + y$ or $-(x + y)$ is $\equiv 1 \pmod 4$. Therefore, there is exactly one generator of $(x + y\sqrt{-1})$ that is primary.

(3) The idea is to mimic Exercise 5.6 with $K = \mathbb{Q}(\zeta_4)$ instead. Without loss of generalities, suppose that $\pi_1 \equiv 1 \pmod 4$.

Let $e_i = 1 - \pi^i$. Note that $2\mathcal{O}_K = (\pi)^2$, and $K_{\mathfrak{p}_2} = \mathbb{Q}_2(\zeta_4)$ with π as its uniformizer. By Exercise 4.4, as $e_{K_{\mathfrak{p}_2}/\mathbb{Q}_2} = 2$, the exponential/log gives an isomorphism

$$(1 + \pi^3\mathcal{O}_{K_{\mathfrak{p}_2}}, \times) \cong (\pi^3\mathcal{O}_{K_{\mathfrak{p}_2}}, +).$$

Note that $\pi^4 = -4$, so $\pi_1 \in 1 + \pi^4\mathcal{O}_{K_{\mathfrak{p}_2}}$. Now note that $\pi_1 = e_4^{a_4}e_5^{a_5}\cdots$ where $a_i \in \{0, 1\}$, just as in Exercise 4.4(3); more precisely, as $\mathcal{O}_{K_{\mathfrak{p}_2}}/\pi\mathcal{O}_{K_{\mathfrak{p}_2}} \cong \mathbb{F}_2$ ($K_{\mathfrak{p}_2}/\mathbb{Q}_2$ is totally ramified), either $\pi_1 \equiv 1 \pmod{\pi^5}$ or $\pi_1 \equiv 1 - \pi^4 \pmod{\pi^5}$, which implies that either π_1 or $\pi_1 e_4^{-1}$ is $\equiv 1 \pmod{\pi^5}$. Continuing this, we get the result. Similarly, $\pi_2 = e_3^{b_3}e_4^{b_4}\cdots$ where $b_i \in \{0, 1\}$.

Note that, for any $x \in 1 + \pi^7\mathcal{O}_{K_{\mathfrak{p}_2}}$, $\log(x) \in \pi^7\mathcal{O}_{K_{\mathfrak{p}_2}}$, so $\frac{\log(x)}{4} \in \pi^3\mathcal{O}_{K_{\mathfrak{p}_2}}$, so $\exp\left(\frac{\log(x)}{4}\right) \in 1 + \pi^3\mathcal{O}_{K_{\mathfrak{p}_2}}$ is well-defined, and $\exp\left(\frac{\log(x)}{4}\right)^4 = \exp(\log(x)) = x$. Therefore, $1 + \pi^7\mathcal{O}_{K_{\mathfrak{p}_2}} \in (\mathcal{O}_{K_{\mathfrak{p}_2}}^\times)^4$. Therefore, $\frac{\pi_1}{e_4^{a_4}e_5^{a_5}e_6^{a_6}}$ is a fourth power, and $\frac{\pi_2}{e_3^{b_3}e_4^{b_4}e_5^{b_5}e_6^{b_6}}$ is a fourth power. Therefore

$$(\pi_1, \pi_2)_{\mathfrak{p}_2} = (e_4^{a_4}e_5^{a_5}e_6^{a_6}, e_3^{b_3}e_4^{b_4}e_5^{b_5}e_6^{b_5})_{\mathfrak{p}_2}.$$

Therefore, it suffices to show that $(e_i, e_j)_{\mathfrak{p}_2} = 1$ for $i + j \geq 7$. Now $e_i + \pi^i e_j = e_{i+j}$. We need to adapt Exercise 5.5 (it cannot be verbatim used as it requires n to be odd);

$$1 = (1 - e_i e_{i+j}^{-1}, e_i e_{i+j}^{-1})_{\mathfrak{p}_2} = (\pi^i e_j e_{i+j}^{-1}, e_i e_{i+j}^{-1})_{\mathfrak{p}_2}.$$

As $e_{i+j} \in 1 + \pi^7\mathcal{O}_{K_{\mathfrak{p}_2}}$, it is a fourth power, so

$$1 = (\pi^i e_j e_{i+j}^{-1}, e_i e_{i+j}^{-1})_{\mathfrak{p}_2} = (\pi^i e_j, e_i)_{\mathfrak{p}_2} = (\pi^i, e_i)_{\mathfrak{p}_2}(e_j, e_i)_{\mathfrak{p}_2}.$$

Note that $e_i = 1 - \pi^i$, so $(\pi^i, e_i)_{\mathfrak{p}_2} = (\pi^i, 1 - \pi^i)_{\mathfrak{p}_2} = 1$, so $(e_j, e_i)_{\mathfrak{p}_2} = 1$, or $(e_i, e_j)_{\mathfrak{p}_2} = 1$, as desired.

(4) This follows from the power reciprocity law and (3) (there is no real embedding of K).

(5) By the earlier discussion and the power reciprocity law, this will follow from $(e_3, e_3)_{\mathfrak{p}_2} = -1$. Note that $e_3 = 1 - \pi^3 = 3 + 2\sqrt{-1}$.

Note that $e_3 \in \mathcal{O}_K$, and for a prime $\mathfrak{p}$ of K, $(e_3, e_3)_{\mathfrak{p}} = 1$ unless $\mathfrak{p}$ divides $4e_3$ (again there is no need to consider archimedean primes as all such primes are complex embeddings and the corresponding Hilbert symbols are trivially 1). Note that $(e_3) = \mathfrak{p}_{13}$ is a prime ideal dividing 13, as $N_{K/\mathbb{Q}}(e_3) = 3^2 + 2^2 = 13$. Thus, letting $\mathfrak{p}_{13} = (e_3)$, by the Hilbert reciprocity law, we have $(e_3, e_3)_{\mathfrak{p}_2} = (e_3, e_3)_{\mathfrak{p}_{13}}$. Now by Theorem 5.44 (tame Hilbert symbols),

$$(e_3, e_3)_{\mathfrak{p}_{13}} = \omega(-1)^{\frac{13-1}{4}} = \omega(-1)^3 = -1,$$

as $f(\mathfrak{p}_{13}|13) = 1$.

(6) This is clearly equivalent to (4) and (5) together.

$\square$